Dimensionality Reduction 2

Introduction to Data Mining

Seoul National University

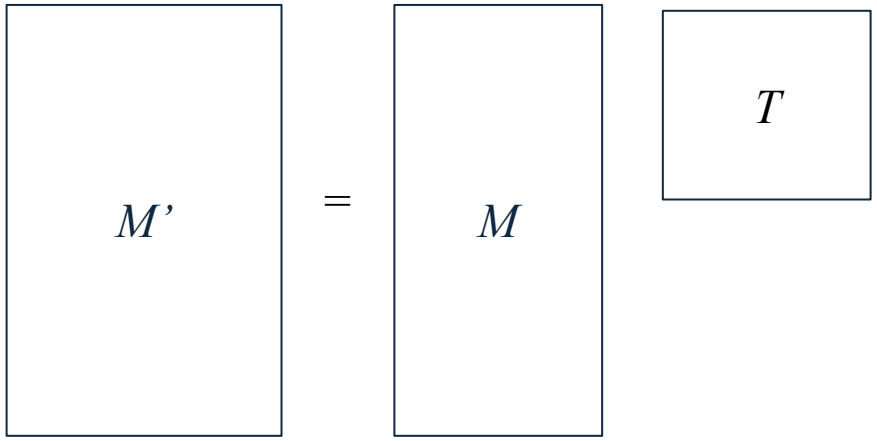
Spring 2026

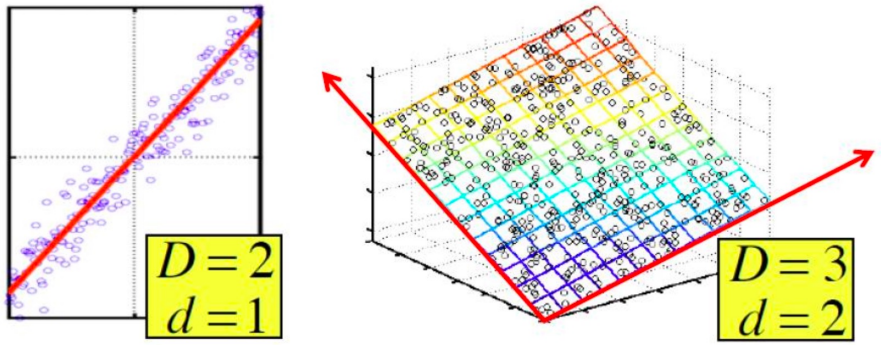
Announcements (Same as Before)

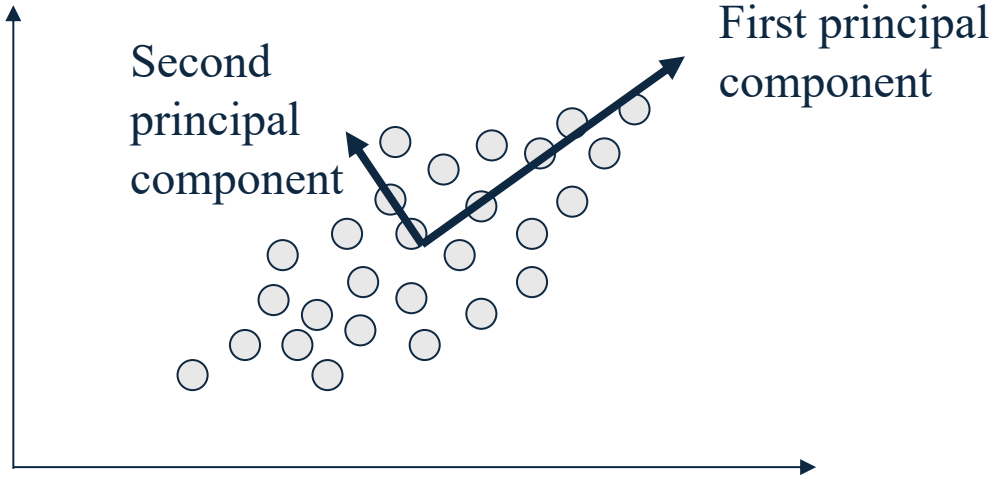
- **Assignments:**
  - HW1 (deadline: April 9)
  - HW2 (deadline: April 30)
- **Midterm:**
  - **When:** April 23 at 9:30 – 10:45
  - **Where:** 302-208 and 302-209 (students will be separated)
  - **How:** Closed-book; no calculators allowed

Recap

- Dimensionality Reduction
- Eigenvalues and Eigenpairs
- Power Iteration
- Principal Component Analysis







Outline

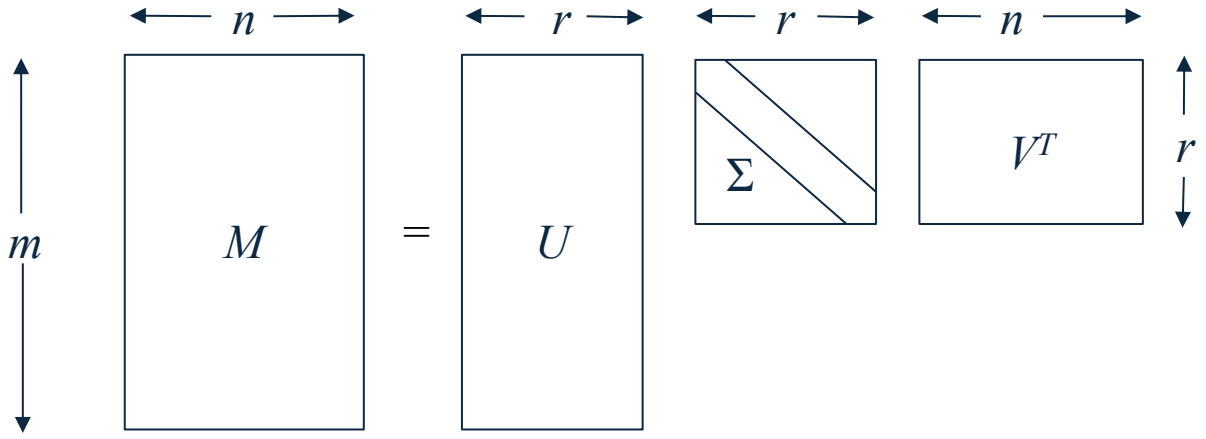
1. **<u>Singular Value Decomposition</u>**
2. Dimensionality Reduction with SVD
3. CUR Decomposition

Singular Value Decomposition

- **Singular value decomposition (SVD):**
  - A factorization of $M \in \mathbb{R}^{m \times n}$ into the product of three matrices:

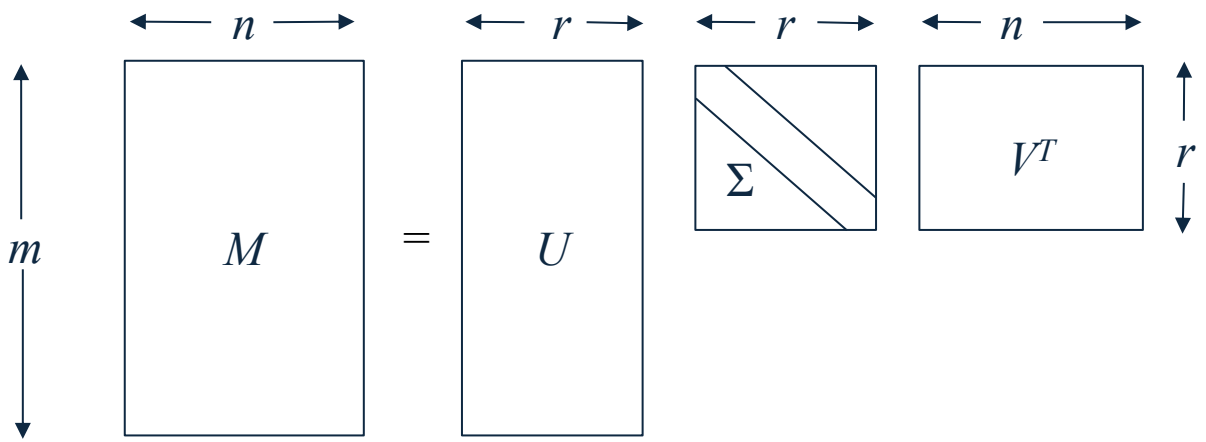
$$M \approx U \Sigma V^{\top},$$

  - whose sizes are determined by an input r.



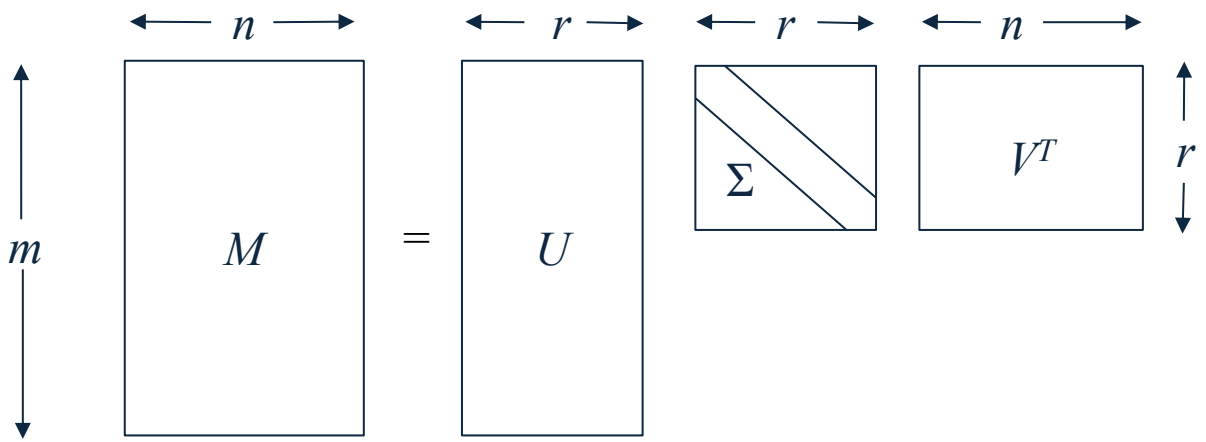
Number of Latent Factors

- r determines the number of **latent factors** to capture.
- Let's define the **reconstruction error** as $\|M - U\Sigma V^\top\|_F$.
  - $\|A\|_F = \mathrm{sqrt}\left(\sum_{ik} a_{ik}^2\right)$ is called the Frobenius norm of a matrix.
- Then, the error reaches zero when $r \geq \mathrm{rank}(M)$.



Properties of Factor Matrices

- U and V are **column-orthonormal matrices** of **singular vectors.**
  - $U^\top U = I$ and $V^\top V = I$, where I is the identity matrix.
- $\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, \cdots, \sigma_r)$ is a **diagonal matrix** of **singular values.**
  - Non-negative and sorted in decreasing order: $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

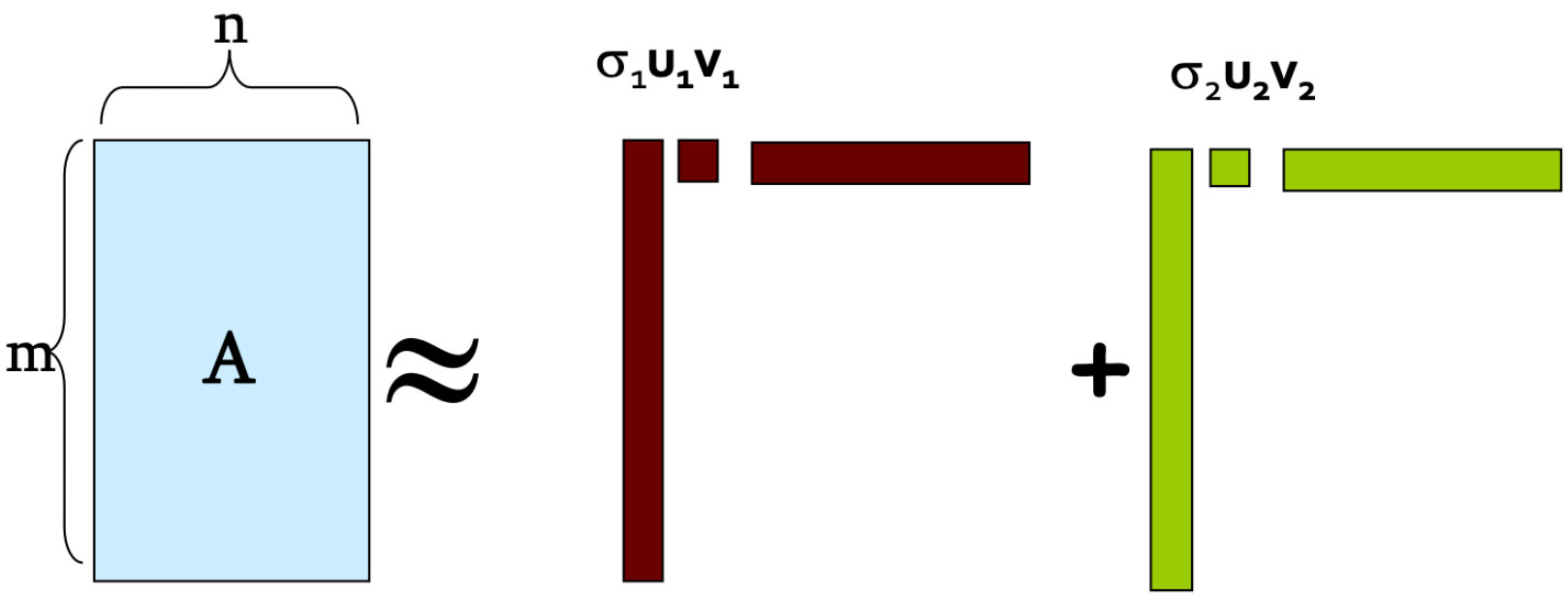


Understanding SVD

- **Property 1:** It is possible to apply SVD to *any* real matrix.
  - Big difference from eigendecomposition.
- **Property 2:** U, Σ and V are unique for each matrix.
- **Property 3:** σ_i denotes the "strength" of the i-th latent factor.
  - Some latent factors are more important than others.
  - Latent factors are normally sorted by their "strength."
    - $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0.$

Different View of SVD

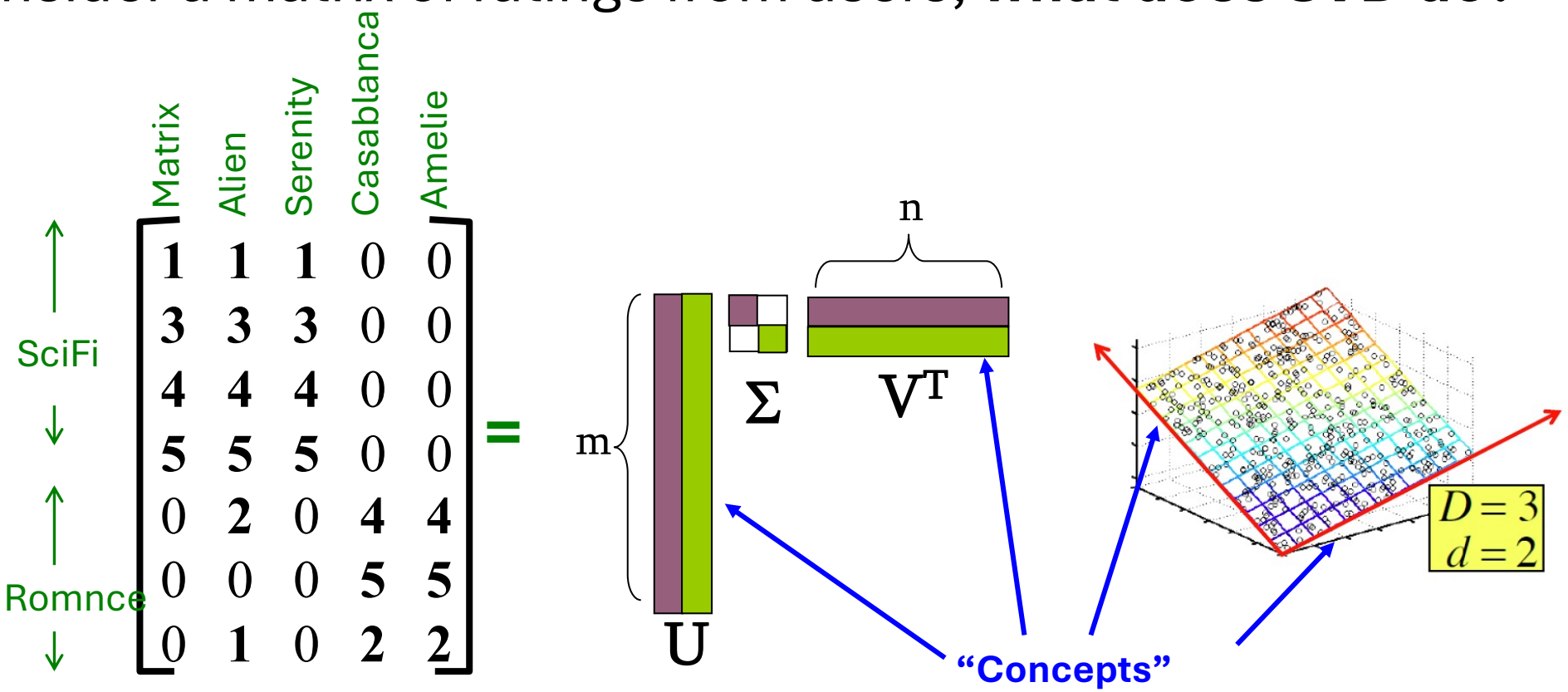
- $M \approx U\Sigma V^\top = \sum_{i=1}^{r} \sigma_i u_i v_i^\top$ (σ_i: scalar, u_i, v_i: singular vectors).
  - SVD is the sum of the r outer products of u_i and v_i vectors.
  - Each $\sigma_i u_i v_i^\top$ represents the effect of the i-th intermediate concept.



Source: Stanford CS246 (2022)

Example: User-Movie Matrix

- Consider a matrix of ratings from users; **what does SVD do?**



"Concepts"
AKA Latent dimensions
AKA Latent factors

Source: Stanford CS246 (2014)

Example: User-Movie Matrix

$$
\begin{array}{c}
\text{SciFi} \\ \uparrow \\ \downarrow \\ \uparrow \\ \text{Romnce} \\ \downarrow
\end{array}
\begin{array}{c}
\begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \end{array} \\
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 2 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 1 & 0 & 2 & 2
\end{bmatrix}
\end{array}
=
\begin{bmatrix}
0.13 & 0.02 & -0.01 \\
0.41 & 0.07 & -0.03 \\
0.55 & 0.09 & -0.04 \\
0.68 & 0.11 & -0.05 \\
0.15 & -0.59 & 0.65 \\
0.07 & -0.73 & -0.67 \\
0.07 & -0.29 & 0.32
\end{bmatrix}
\times
\begin{bmatrix}
12.4 & 0 & 0 \\
0 & 9.5 & 0 \\
0 & 0 & 1.3
\end{bmatrix}
\times
\begin{bmatrix}
0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
0.40 & -0.80 & 0.40 & 0.09 & 0.09
\end{bmatrix}
$$

"strength" of the SciFi-concept

Example

U **is "user-to-concept" similarity matrix**

$$
\begin{array}{c}
\\ \text{SciFi} \\ \\ \\ \\ \text{Romnce} \\ \\
\end{array}
\begin{array}{c}
\begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \end{array} \\
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 2 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 1 & 0 & 2 & 2
\end{bmatrix}
\end{array}
=
\begin{array}{c}
\begin{array}{ccc} \text{SciFi-concept} & \text{Romance-concept} & \\ \end{array} \\
\begin{bmatrix}
0.13 & 0.02 & -0.01 \\
0.41 & 0.07 & -0.03 \\
0.55 & 0.09 & -0.04 \\
0.68 & 0.11 & -0.05 \\
0.15 & -0.59 & 0.65 \\
0.07 & -0.73 & -0.67 \\
0.07 & -0.29 & 0.32
\end{bmatrix}
\end{array}
\times
\begin{bmatrix}
12.4 & 0 & 0 \\
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0 & 0 & 1.3
\end{bmatrix}
\times
\begin{bmatrix}
0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
0.40 & -0.80 & 0.40 & 0.09 & 0.09
\end{bmatrix}
$$

Example: User-Movie Matrix

V **is "movie-to-concept" similarity matrix**

SciFi-concept

SciFi

Romnce

$$
\begin{array}{c}
\begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \end{array} \\
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 2 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 1 & 0 & 2 & 2
\end{bmatrix}
\end{array}
=
\begin{bmatrix}
0.13 & 0.02 & -0.01 \\
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0.55 & 0.09 & -0.04 \\
0.68 & 0.11 & -0.05 \\
0.15 & -0.59 & 0.65 \\
0.07 & -0.73 & -0.67 \\
0.07 & -0.29 & 0.32
\end{bmatrix}
\times
\begin{bmatrix}
12.4 & 0 & 0 \\
0 & 9.5 & 0 \\
0 & 0 & 1.3
\end{bmatrix}
\times
\begin{bmatrix}
0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
0.40 & -0.80 & 0.40 & 0.09 & 0.09
\end{bmatrix}
$$

SciFi-concept

Example: User-Movie Matrix

$$
\begin{array}{c}
\text{SciFi} \\ \uparrow \\ \downarrow \\ \\ \uparrow \\ \text{Romnce} \\ \downarrow
\end{array}
\begin{array}{c}
\begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \end{array} \\
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 2 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 1 & 0 & 2 & 2
\end{bmatrix}
\end{array}
=
\begin{array}{c}
\text{What concept?} \\
\downarrow \\
\begin{bmatrix}
0.13 & 0.02 & -0.01 \\
0.41 & 0.07 & -0.03 \\
0.55 & 0.09 & -0.04 \\
0.68 & 0.11 & -0.05 \\
0.15 & -0.59 & 0.65 \\
0.07 & -0.73 & -0.67 \\
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\end{bmatrix}
\end{array}
\times
\begin{bmatrix}
12.4 & 0 & 0 \\
0 & 9.5 & 0 \\
0 & 0 & 1.3
\end{bmatrix}
\times
\begin{bmatrix}
0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
0.40 & -0.80 & 0.40 & 0.09 & 0.09
\end{bmatrix}
$$

Example: User-Movie Matrix

- **Summary:** SVD provides **interpretable decomposition.**
  - U: User (row)-to-concept matrix.
  - V: Movie (column)-to-concept matrix.
  - Σ: Its diagonal elements mean the “strength” of each concept.

Computing SVD: Step 1

- **Q:** How can we actually find U, Σ, and V?
- We start by assuming $M = U\Sigma V^\top$.
- $M^\top = (U\Sigma V^\top)^\top = (V^\top)^\top \Sigma^\top U^\top = V\Sigma^\top U^\top$.
- $M^\top M = V\Sigma^\top U^\top U\Sigma V^\top = V\Sigma^\top \Sigma V^\top = V\Sigma^2 V^\top$.
  - U is column-orthonormal; $U^\top U = I$.
  - Σ is diagonal; $\Sigma^\top = \Sigma$.
- $M^\top MV = V\Sigma^2 V^\top V = V\Sigma^2$.
  - V is also column-orthonormal.

Computing SVD: Step 2

- From $M^\top MV = V\Sigma^2$, let's focus on the i-th column of V.
- Then, v_i is an eigenvector of $M^\top M$, since $M^\top M v_i = \sigma^2 v_i$.
  - The corresponding eigenvalue is $\sigma^2 = \Sigma_{ii}^2$.
- Thus, we can find V and Σ through the eigenpairs of $M^\top M$.
  - Singular values are the square root of the eigenvalues.
  - Similarly, $MM^\top$ gives U (and Σ).

Connecting SVD and PCA

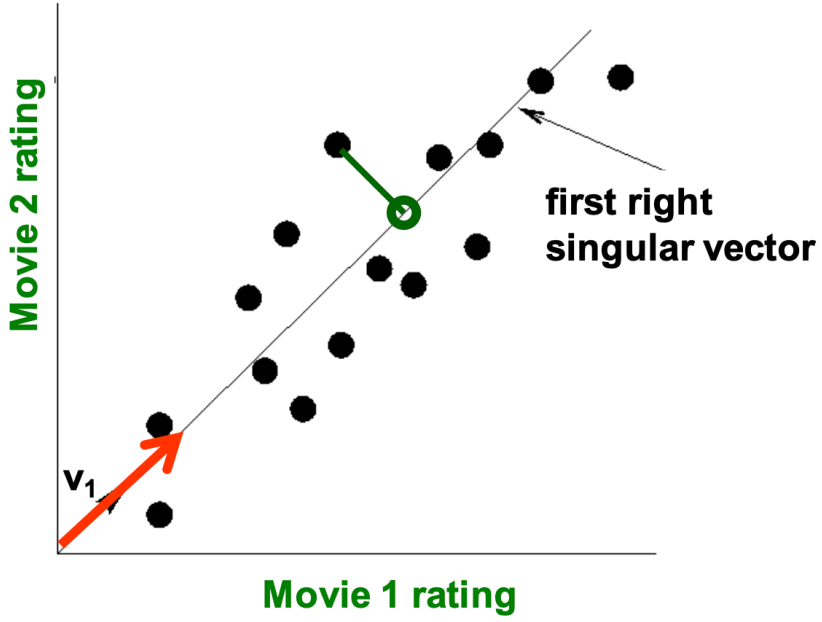
- **Right singular vectors** in V = **Principal components** of PCA.
  - **Note:** PCA needs column-centering, while SVD doesn't care.
- In reality, the full computation of PCA is slow.
  - Creating $M^\top M$ is very slow If M is large (e.g., billions of rows).
  - Eigendecomposition of $M^\top M$ is even slower.
- Thus, SVD is a standard way of computing PCA.
  - Efficient algorithms exist for computing SVD without creating $M^\top M$.

Outline

1. Singular Value Decomposition
2. **<u>Dimensionality Reduction with SVD</u>**
3. CUR Decomposition

SVD for Dimensionality Reduction

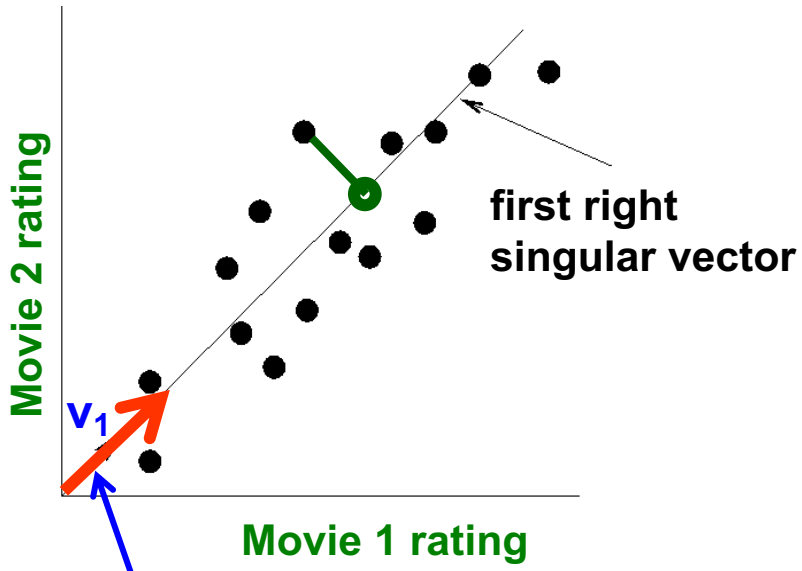
- Let's use fewer coordinates to describe point positions.
  - Consider an example of N users for 2 movies of 1 concept.
  - Point's position can be described as its location along vector v_1.



Source: Stanford CS246 (2022)

Example: SVD

- U is about users, V is about movies.
- Each column of V is a **new axis**.

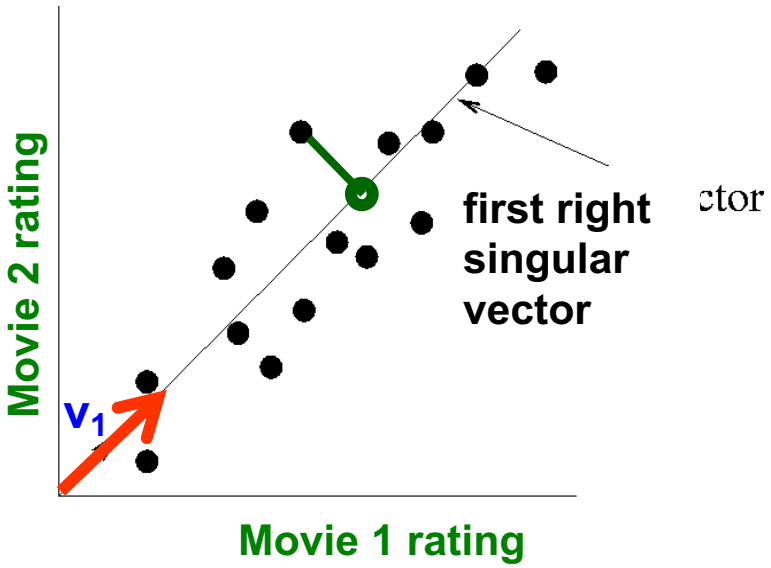


$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \times \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \times \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

$$M = U\Sigma V^{\top}$$

Example: SVD

- First column of V:
  - The largest variance.



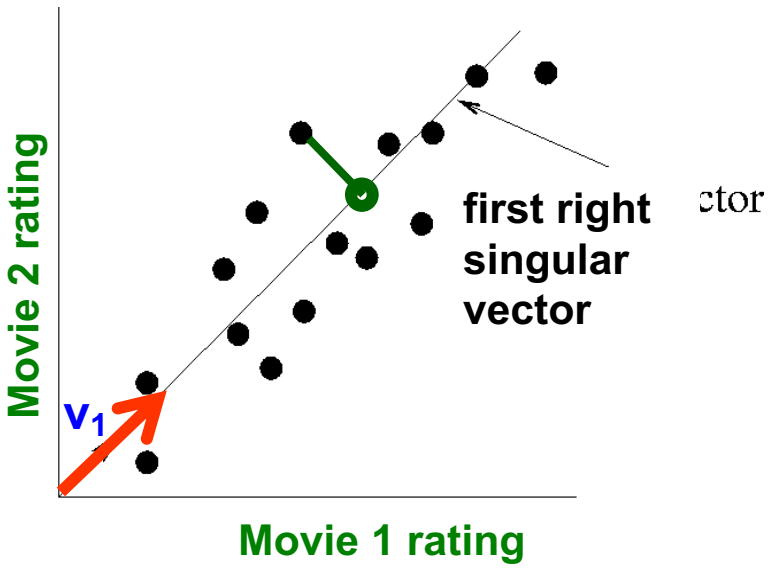
variance ('spread') on the v_1 axis

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0.13 & 0.02 & -0.01 \\ 0.41 & 0.07 & -0.03 \\ 0.55 & 0.09 & -0.04 \\ 0.68 & 0.11 & -0.05 \\ 0.15 & -0.59 & 0.65 \\ 0.07 & -0.73 & -0.67 \\ 0.07 & -0.29 & 0.32 \end{bmatrix} \times \begin{bmatrix} 12.4 & 0 & 0 \\ 0 & 9.5 & 0 \\ 0 & 0 & 1.3 \end{bmatrix} \times \begin{bmatrix} 0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\ 0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\ 0.40 & -0.80 & 0.40 & 0.09 & 0.09 \end{bmatrix}$$

$$M = U\Sigma V^{\top}$$

Example: SVD

- $U\Sigma$ gives the points coordinates.
  - In the projection axis.



$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix}$$

Projection of users on the "Sci-Fi" axis
$(U\,\Sigma)^T$:

$$\begin{bmatrix} 1.61 & 0.19 & -0.01 \\ 5.08 & 0.66 & -0.03 \\ 6.82 & 0.85 & -0.05 \\ 8.43 & 1.04 & -0.06 \\ 1.86 & -5.60 & 0.84 \\ 0.86 & -6.93 & -0.87 \\ 0.86 & -2.75 & 0.41 \end{bmatrix}$$

$$M = U\Sigma V^{\top}$$

SVD for Dimensionality Reduction

- **Q:** How exactly is dimensionality reduction done by SVD?
- **A:** Set smallest singular values to zero.

$$
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 2 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 1 & 0 & 2 & 2
\end{bmatrix}
\approx
\begin{bmatrix}
0.13 & 0.02 & -0.01 \\
0.41 & 0.07 & -0.03 \\
0.55 & 0.09 & -0.04 \\
0.68 & 0.11 & -0.05 \\
0.15 & -0.59 & 0.65 \\
0.07 & -0.73 & -0.67 \\
0.07 & -0.29 & 0.32
\end{bmatrix}
\times
\begin{bmatrix}
12.4 & 0 & 0 \\
0 & 9.5 & 0 \\
0 & 0 & 1.3
\end{bmatrix}
\times
\begin{bmatrix}
0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
0.40 & -0.80 & 0.40 & 0.09 & 0.09
\end{bmatrix}
$$

SVD for Dimensionality Reduction

- Reconstruction error is quantified by the **Frobenius norm**:
  - In this case, we have $\|M - B\|_F^2 = \sum_{ik}(M_{ik} - B_{ik})^2$.

$$\begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} & 0 & 0 \\ \mathbf{3} & \mathbf{3} & \mathbf{3} & 0 & 0 \\ \mathbf{4} & \mathbf{4} & \mathbf{4} & 0 & 0 \\ \mathbf{5} & \mathbf{5} & \mathbf{5} & 0 & 0 \\ 0 & \mathbf{2} & 0 & \mathbf{4} & \mathbf{4} \\ 0 & 0 & 0 & \mathbf{5} & \mathbf{5} \\ 0 & \mathbf{1} & 0 & \mathbf{2} & \mathbf{2} \end{bmatrix} \approx \begin{bmatrix} \mathbf{0.92} & \mathbf{0.95} & \mathbf{0.92} & 0.01 & 0.01 \\ \mathbf{2.91} & \mathbf{3.01} & \mathbf{2.91} & -0.01 & -0.01 \\ \mathbf{3.90} & \mathbf{4.04} & \mathbf{3.90} & 0.01 & 0.01 \\ \mathbf{4.82} & \mathbf{5.00} & \mathbf{4.82} & 0.03 & 0.03 \\ 0.70 & \mathbf{0.53} & 0.70 & \mathbf{4.11} & \mathbf{4.11} \\ -0.69 & 1.34 & -0.69 & \mathbf{4.78} & \mathbf{4.78} \\ 0.32 & \mathbf{0.23} & 0.32 & \mathbf{2.01} & \mathbf{2.01} \end{bmatrix}$$

Reconstructed data matrix B.

Best Low-Rank Approximation

- Given $r < \mathrm{rank}(M)$, SVD gives the **best axes** to project on.
  - In other words, it gives the **best rank-r approximation** of the matrix.
  - That minimizes the reconstruction error $\|M - B\|_F$.
- **Intuitive reason:**
  - σ_i represents how much "energy" the matrix has along v_i.
  - Truncation keeps the top r orthogonal "energy directions."
  - As a result, SVD can minimize the loss of energy.

Determining the Low Rank

- **Q:** What is a good value for r?
- **A:** Pick r to have at least 90% of the **total energy.**
  - We define **energy** to be the sum of squared singular values.
- **Back to our example:**
  - Singular values are 12.4, 9.5, and 1.3 (total energy = 245.7).
  - If we drop 1.3, whose square is 1.7, the remaining is energy 244.0.
  - The remaining energy is over 99% of the total energy.

Querying Using Concepts

- **Q:** Given the result of SVD, what movies would a new user like?

$$
\begin{array}{c}
\text{SciFi} \\
\uparrow \\
\downarrow \\
\uparrow \\
\text{Romnce} \\
\downarrow
\end{array}
\begin{array}{c}
\begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \end{array} \\
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 2 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
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\end{array}
=
\begin{bmatrix}
0.13 & 0.02 & -0.01 \\
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\begin{bmatrix}
0.56 & 0.59 & 0.56 & 0.09 & 0.09 \\
0.12 & -0.02 & 0.12 & -0.69 & -0.69 \\
0.40 & -0.80 & 0.40 & 0.09 & 0.09
\end{bmatrix}
$$

Querying Using Concepts

- Assume that the new user is represented as $q = [5 \;\; 0 \;\; 0 \;\; 0 \;\; 0]$.
  - Rated Matrix with the score 5.
- **Step 1:** Multiply by V to map q into the concept space as qV:

$$\mathbf{q} = \begin{array}{c} \begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \end{array} \\ \begin{bmatrix} 5 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \mathbf{x} \begin{bmatrix} 0.56 & 0.12 \\ 0.59 & -0.02 \\ 0.56 & 0.12 \\ 0.09 & -0.69 \\ 0.09 & -0.69 \end{bmatrix} = \begin{array}{c} \begin{array}{cc} \text{SciFi-concept} & \end{array} \\ \begin{bmatrix} 2.8 & 0.6 \end{bmatrix} \end{array}$$

movie-to-concept similarities (V)

Querying Using Concepts

- **Step 2:** Multiply by $V^\top$ back to the movie space to get:

$$qVV^\top = [1.64 \;\; 1.64 \;\; 1.64 \;\; -0.162 \;\; -0.162]$$

- He/she would probably like Alien and Serenity.
  - Sci-Fi movies.

$$\begin{array}{c} \\ \uparrow \\ \text{SciFi} \\ \downarrow \\ \uparrow \\ \text{Romnce} \\ \downarrow \end{array} \begin{array}{c} \begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Serenity} & \text{Casablanca} & \text{Amelie} \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 2 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 1 & 0 & 2 & 2 \end{bmatrix} \end{array}$$

Querying Using Concepts

- **Alternate query:** Find users who are similar to the target user q.
  - User d is similar to user q even without any ratings in common.

	Matrix	Alien	Serenity	Casablanca	Amelie		SciFi-concept
d =	0	4	5	0	0	→	5.2 0.4
q =	5	0	0	0	0	→	2.8 0.6

Zero ratings in common

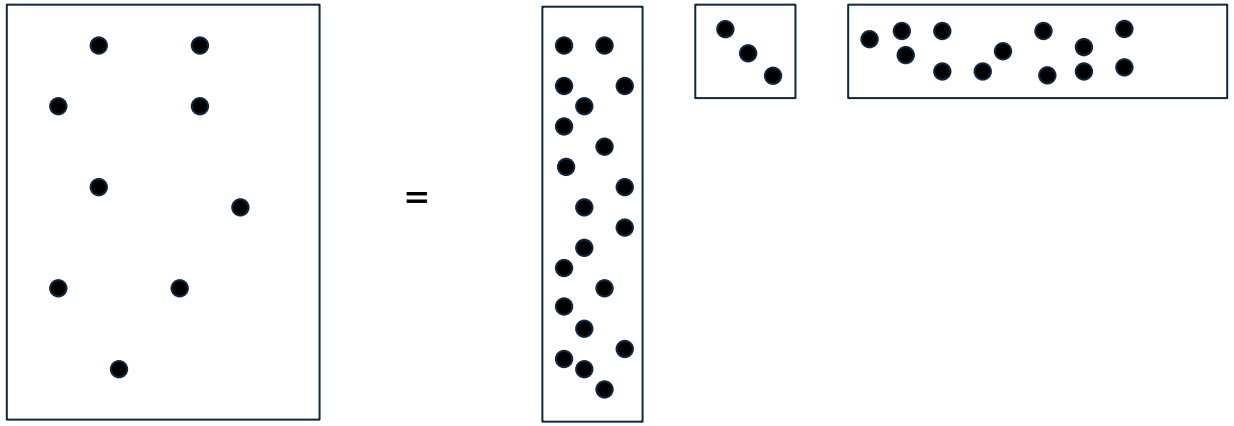
Similarity ≠ 0

Outline

1. Singular Value Decomposition
2. Dimensionality Reduction with SVD
3. **<u>CUR Decomposition</u>**

Sparsity of a Data Matrix

- It is common for the data matrix M to be highly sparse.
  - E.g., a matrix of customers and products.
- However, U and V from SVD will not be sparse.
  - Σ is sparse, but much smaller than U and V, so it does not help.



CUR Decomposition

- **Goal:** Express $M \approx CUR$ while minimizing $\|M - CUR\|_F$.
- **CUR decomposition** leverages the sparsity of M by:
  - Using (sparse) rows and columns of M for creating C and R.
  - Only the matrix U (analogous to Σ) is dense, but it is small.
- CUR is always an approximation unlike SVD.
  - No matter how large r (= number of concepts) is.
  - Still, there is a theory that CUR converges to M as r gets larger.

Definition of CUR

- A CUR decomposition of an $m \times n$ matrix M is:
  - $m \times r$ matrix C, a randomly chosen set of r columns of M.
  - $r \times n$ matrix R, a randomly chosen set of r rows of M.
  - $r \times r$ matrix U constructed from C and R to minimize the error.

$$
\begin{array}{r}
\text{Joe} \\ \text{Jim} \\ \text{John} \\ \text{Jack} \\ \text{Jill} \\ \text{Jenny} \\ \text{Jane} \\ \\
\end{array}
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 0 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 0 & 0 & 2 & 2
\end{bmatrix}
\approx
\overset{C}{\begin{bmatrix}
1.54 & 0 \\
4.63 & 0 \\
6.17 & 0 \\
7.72 & 0 \\
0 & 9.30 \\
0 & 11.63 \\
0 & 4.65
\end{bmatrix}}
\overset{U}{\begin{bmatrix}
0 & 1/25 \\
1/25 & 0
\end{bmatrix}}
\overset{R}{\begin{bmatrix}
0 & 0 & 0 & 7.79 & 7.79 \\
6.36 & 6.36 & 6.36 & 0 & 0
\end{bmatrix}}
$$

Sampled rows and columns are scaled.

Weighted Sampling of Rows/Columns

- Uniform sampling of rows or columns is not optimal.
  - Some are more important for reducing the reconstruction error.
- We measure the importance by the Frobenius norm.
  - Let $f = \|M\|_F^2$ be the square of M's Frobenius norm.
  - Row i is selected with a probability $p_i = \sum_k m_{ik}^2 \,/\, f$.
  - Column k is selected with a probability $q_k = \sum_i m_{ik}^2 \,/\, f$.

Scaling of Rows/Columns

- After selecting each column k, divide each element by $\sqrt{rq_k}$.
  - $\sqrt{r}$ to unify the scale across different choices of r.
  - $\sqrt{q_k}$ to unify the scale between differently important columns.
- Do the analogous for each selected row.
- Same rows or columns can be selected more than once.
  - Not a big problem though.

Example: Choosing a Row

- The (squared) Frobenius norm of M is 243.
- The probability to choose Jack is $75/243 = 0.309$.
  - Jack has the Frobenius norm of $\sqrt{75}$.
- The selected row is divided by $\sqrt{rp} = \sqrt{2 \times 0.309}$.
  - If $r = 2$.
- We have $[6.36 \;\; 6.36 \;\; 6.36 \;\; 0 \;\; 0]$ as a row of R.

$$
\begin{array}{r c}
 & \begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Star Wars} & \text{Casablanca} & \text{Titanic} \end{array} \\
\begin{array}{r} \text{Joe} \\ \text{Jim} \\ \text{John} \\ \text{Jack} \\ \text{Jill} \\ \text{Jenny} \\ \text{Jane} \end{array} &
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 0 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 0 & 0 & 2 & 2
\end{bmatrix}
\end{array}
$$

Constructing the Middle Matrix

- Let W be the $r \times r$ matrix that is the "intersection" of C and R.
- Compute the SVD of W, say, $W = X\Sigma Y^\top$.
- Compute Σ^+, the **Moore-Penrose pseudoinverse** of Σ.
  - If the i-th diagonal element of Σ is $\sigma \neq 0$, replace it by $1/\sigma$.
  - Otherwise, leave it as 0.
  - Identical to the inverse if the matrix is invertible.
- Return $U = Y(\Sigma^+)^2 X^\top$.
  - Carefully constructed so that the CUR decomposition approximates M.

Example: Middle Matrix

- Construct the intersection matrix W as follows:
  - **Columns:** Alien & Casablanca.
  - **Rows:** Jenny & Jack.
- The order affects the intersection.
  - Jenny & Jack ≠ Jack & Jenny.

$$
\begin{array}{r}
\\ \text{Joe} \\ \text{Jim} \\ \text{John} \\ \text{Jack} \\ \text{Jill} \\ \text{Jenny} \\ \text{Jane}
\end{array}
\begin{array}{c}
\begin{array}{ccccc} \text{Matrix} & \text{Alien} & \text{Star Wars} & \text{Casablanca} & \text{Titanic} \end{array} \\
\begin{bmatrix}
1 & 1 & 1 & 0 & 0 \\
3 & 3 & 3 & 0 & 0 \\
4 & 4 & 4 & 0 & 0 \\
5 & 5 & 5 & 0 & 0 \\
0 & 0 & 0 & 4 & 4 \\
0 & 0 & 0 & 5 & 5 \\
0 & 0 & 0 & 2 & 2
\end{bmatrix}
\end{array}
\qquad
W = \begin{bmatrix} 0 & 5 \\ 5 & 0 \end{bmatrix}
$$

Example: Middle Matrix

- Construct U from W as follows:
  - SVD for W is $X\Sigma Y^\top = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 5 \\ 5 & 0 \end{bmatrix}$.
  - $\Sigma^+ = \begin{bmatrix} 1/5 & 0 \\ 0 & 1/5 \end{bmatrix}$.
  - $U = Y(\Sigma^+)^2 X^\top = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/5 & 0 \\ 0 & 1/5 \end{bmatrix}^2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1/25 \\ 1/25 & 0 \end{bmatrix}$.

Complete CUR Decomposition

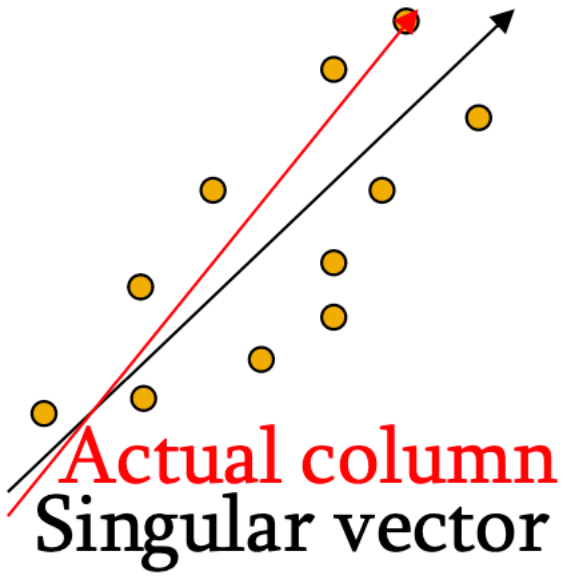
- We now have a method to select C, U, and R:
  - Note that CUR randomly creates C and R.
  - Then, U is created deterministically from C and R.

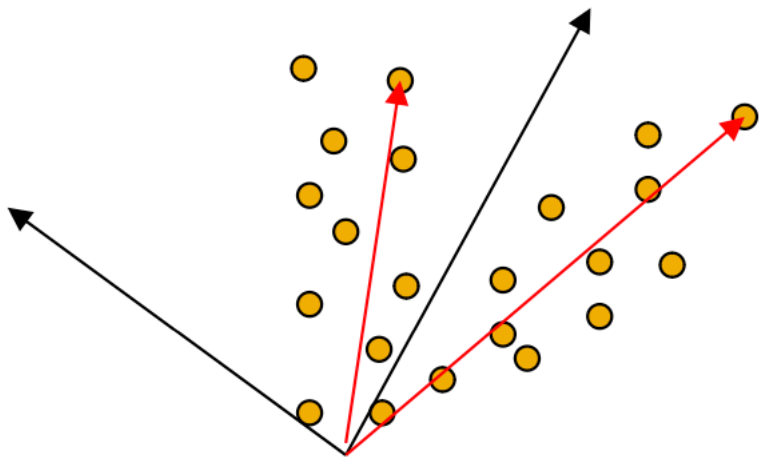
$$\begin{array}{r} \text{Joe} \\ \text{Jim} \\ \text{John} \\ \text{Jack} \\ \text{Jill} \\ \text{Jenny} \\ \text{Jane} \end{array} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 3 & 3 & 3 & 0 & 0 \\ 4 & 4 & 4 & 0 & 0 \\ 5 & 5 & 5 & 0 & 0 \\ 0 & 0 & 0 & 4 & 4 \\ 0 & 0 & 0 & 5 & 5 \\ 0 & 0 & 0 & 2 & 2 \end{bmatrix}$$

$$CUR = \begin{bmatrix} 1.54 & 0 \\ 4.63 & 0 \\ 6.17 & 0 \\ 7.72 & 0 \\ 0 & 6.58 \\ 0 & 8.22 \\ 0 & 3.29 \end{bmatrix} \begin{bmatrix} 0 & 1/25 \\ 1/25 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 7.79 & 7.79 \\ 6.36 & 6.36 & 6.36 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.39 & 0.39 & 0.39 & 0 & 0 \\ 1.18 & 1.18 & 1.18 & 0 & 0 \\ 1.57 & 1.57 & 1.57 & 0 & 0 \\ 1.96 & 1.96 & 1.96 & 0 & 0 \\ 0 & 0 & 0 & 2.05 & 2.05 \\ 0 & 0 & 0 & 2.56 & 2.56 \\ 0 & 0 & 0 & 1.02 & 1.02 \end{bmatrix}$$

CUR vs. SVD

- CUR is like selecting representative points (like clustroids).
- SVD is like creating new representative directions (like centroids).
- They work differently if two point clouds exist at different angles.





Outline

1. Singular Vector Decomposition
   - Singular values and vectors
   - Relationship to PCA
2. Dimensionality Reduction with SVD
   - Best low-rank approximation
3. CUR Decomposition
   - Advantage in a sparse data matrix